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UNE-EN IEC 62402:2019

Idioma: Inglés

Gestión de la obsolescencia. (Ratificada por la Asociación Española de Normalización en agosto de 2019.)

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UNE-EN IEC 62402:2019

Gestión de la obsolescencia. (Ratificada por la Asociación Española de Normalización en agosto de 2019.)

Obsolescence management (Endorsed by Asociación Española de Normalización in August of 2019.)

Gestion de l'obsolescence (Entérinée par l'Asociación Española de Normalización en août 2019.)

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EUROPEAN STANDARD

NORME EUROPÉENNE

EUROPÄISCHE NORM

EN IEC 62402

July 2019

ICS 21.020

Supersedes EN 62402:2007 and all of its amendments and corrigenda (if any)

English Version

Obsolescence management (IEC 62402:2019)

Gestion de l'obsolescence
(IEC 62402:2019)

Obsoleszenzmanagement
(IEC 62402:2019)

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European foreword

The text of document 56/1838/FDIS, future edition 2 of IEC 62402, prepared by IEC/TC 56 "Dependability" was submitted to the IEC-CENELEC parallel vote and approved by CENELEC as EN IEC 62402:2019.

The following dates are fixed:

• latest date by which the document has to be implemented at national level by publication of an identical national standard or by endorsement	(dop)	2020-04-03
• latest date by which the national standards conflicting with the document have to be withdrawn	(dow)	2022-07-03

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IEC 60300-1	NOTE	Harmonized as EN 60300-1
IEC 62239-1	NOTE	Harmonized as EN IEC 62239-1
IEC/ISO 31010	NOTE	Harmonized as EN 31010
IEC 62474	NOTE	Harmonized as EN IEC 62474
IEC 62668-1	NOTE	Harmonized as EN 62668-1[1]
IEC 62435-1	NOTE	Harmonized as EN 62435-1
IEC 62435-4	NOTE	Harmonized as EN IEC 62435-4
IEC 62668-2	NOTE	Harmonized as EN IEC 62668-2[2]

[1] Under preparation. Stage at the time of publication: prEN 62668-1
[2] Under preparation. Stage at the time of publication: FprEN IEC 62668-2

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IEC 62402

Edition 2.0 2019-05

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Obsolescence management

Gestion de l'obsolescence

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IEC 62402

Edition 2.0 2019-05

INTERNATIONAL STANDARD

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Obsolescence management

Gestion de l'obsolescence

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ELECTROTECHNICAL
COMMISSION

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ICS 21.020

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CONTENTS

FOREWORD 5
INTRODUCTION 7
1 Scope 8
2 Normative references 8
3 Terms, definitions and abbreviated terms 8
3.1 Terms and definitions 8
3.2 Abbreviated terms 11
4 Obsolescence management 12
4.1 What is obsolescence 12
4.2 What is obsolescence management 13
4.3 Benefits of obsolescence management 13
4.4 Obsolescence management process 14
5 Obsolescence management policy 17
6 Infrastructure and organization 18
6.1 General 18
6.2 Management responsibilities 18
6.3 Obsolescence management organization 18
6.4 Customer/manufacturer management 18
6.5 Partnering agreements between organizations 19
7 Development of an OMP 19
7.1 OMP 19
7.2 OMP contents 19
8 Strategies to minimize obsolescence during design 21
8.1 Obsolescence as a consideration in design 21
8.2 Source code 21
8.3 Material characterization 21
8.4 Modularity 21
8.5 Transparency 21
8.6 Sustainable technologies including materials 22
8.7 Open standards 22
8.8 Obtaining IPR 22
8.9 Software licensing 22
8.10 Data acquisition 22
9 Obsolescence management approach 23
9.1 Introduction to risk assessment 23
9.2 Obsolescence monitoring 24
9.2.1 Monitoring background 24
9.2.2 Obsolescence notice monitoring 25
9.2.3 Direct contact monitoring 25
9.3 Risk assessment to select approach 25
9.4 Proactive approach 27
9.5 Reactive approach 27
10 Obsolescence resolutions 27
10.1 Resolution selection and implementation 27

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10.2 Same item 28
10.3 Life of need buy 29
10.4 Substitutes 30
10.5 Emulation and reverse engineering 30
10.6 Design change 31
11 Measurement and improvement of obsolescence management activities 31
11.1 General 31
11.2 Metrics 31
Annex A (informative) Vocabulary relating to obsolescence 33
Annex B (informative) Obsolescence resolutions 36
B.1 Obsolescence management resolutions with EOP forecast 36
B.2 Obsolescence management resolutions with EOP announcement 37
B.3 Alternate manufacturers: example of a reactive approach in electronics 38
Annex C (informative) Guidance on the effects of obsolescence 39
Annex D (informative) Guidance on the OMP 40
Annex E (informative) Examples of an obsolescence risk assessment 41
E.1 General 41
E.2 Example 1 41
E.3 Example 2 42
E.3.1 Risk assessment process 42
E.3.2 Likelihood assessment 43
E.3.3 Impact date assessment 43
E.3.4 Obsolescence risk 44
E.3.5 Review 44
Annex F (informative) Example of an obsolescence management decision process 45
Bibliography 47

Figure 1 – Obsolescence management activities 14
Figure 2 – Assessments identifying obsolescence risks and issues 15
Figure 3 – Item's life cycle versus obsolescence management activities 17
Figure 4 – Proactive versus reactive approaches with resolutions 24
Figure B.1 – Item production output with EOP forecast and obsolescence scenarios 36
Figure B.2 – Item production output at EOP announcement and obsolescence scenarios 37
Figure E.1 – Sample risk assessment process 43
Figure F.1 – Initial decision process to recommend obsolescence resolutions 45
Figure F.2 – Decision process to recommend obsolescence resolutions (long term repairs strategy) 46
Figure F.3 – Decision process to recommend obsolescence resolutions (LNB) 46

Table E.1 – Likelihood assessment 41
Table E.2 – Impact assessment 41
Table E.3 – Combination of likelihood and impact assessment 42

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Table E.4 – Level of proactive approach assessment 42
Table E.5 – EOP forecast 43
Table E.6 – Number of approved manufacturers 43
Table E.7 – Likelihood 43
Table E.8 – Impact date 44
Table E.9 – Risk level 44
Table E.10 – Obsolescence risk review 44

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OBSOLESCENCE MANAGEMENT

FOREWORD

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This second edition cancels and replaces the first edition published in 2007. This edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

a) this document has now been written with requirements as a standard, not a guide;
b) this document continues to have guidance in the informative annexes;
c) this document has been written as a general process for all technologies and items.

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The text of this International Standard is based on the following documents:

FDIS	Report on voting
56/1838/FDIS	56/1843/RVD

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INTRODUCTION

For the purposes of this document, obsolescence management is a discipline used at all phases of an item's life cycle to ensure an item and its sub items can continue to fulfil their requirements over their expected useful life.

This document takes a different view of obsolescence from the standard dictionary definition. Instead of an item becoming outdated or no longer used, this document views obsolescence as the transition of a required item still in use from available to unavailable from the manufacturer. Any item that remains in use will be subject to obsolescence. Obsolescence manifests itself as difficulty in obtaining supplies, spares and/or support.

This document defines the requirements for managing the obsolescence of any type of item. Obsolescence management helps prevent unnecessary losses (for example loss of commercial service or capability) and treat risks associated with obsolescence. The assessment of risk associated with obsolescence takes account of factors including but not limited to: the likelihood of an item becoming obsolete during its expected useful life, the likelihood of an impact occurring during that projected useful life, and the severity of that impact. Obsolescence management treats risks associated with obsolescence by reducing the likelihood or severity of impact, or both.

It has become essential to include obsolescence management within planning activities from the earliest life cycle phases. The guidance provided in this document could be characterized as strategic obsolescence management when obsolescence management is planned and implemented during the early life cycle phases.

Even though this situation may not be a direct case of obsolescence, this document will also be of assistance in the management of items that have diminished manufacturing sources and materiel shortages that can result in long lead times, reduced availability and ultimately obsolescence of those items.

Managing obsolescence contributes to the dependability of an item, particularly supportability, which is defined as the 'ability to be supported to sustain the required availability with a defined operational profile and given logistic and maintenance resources'. As such, obsolescence management may be performed as part of an overall dependability management programme as described in IEC 60300-1 [1][1].

1 Numbers in square brackets refer to the Bibliography.

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OBSOLESCENCE MANAGEMENT

1 Scope

This document provides requirements and guidance for obsolescence management applicable to any organization that is dependent on another organization to obtain value from the usefulness of the items that it provides. A cost-effective obsolescence management process and the activities used to implement the process are applicable throughout all phases of an item's life cycle.

This document covers the following areas:

- establishing an obsolescence management policy;
- establishing an infrastructure and an organization;
- developing an obsolescence management plan (OMP);
- developing strategies to minimize obsolescence during design;
- determining an obsolescence management approach;
- selecting obsolescence resolution and implementation;
- measuring and improving the performance of the outcomes of the obsolescence management activities.

Guidance on obsolescence management is included as notes, in the informative annexes and references in the Bibliography.

2 Normative references

There are no normative references in this document.

3 Terms, definitions and abbreviated terms

3.1 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at http://www.electropedia.org/
- ISO Online browsing platform: available at http://www.iso.org/obp

3.1.1
alternative item
item whose characteristics can be different from that specified for one or more reasons

EXAMPLE Items with different quality or reliability level, tolerance, parameteters, temperature range.

Note 1 to entry: See also 'substitutes' (10.4).

3.1.2
commercial off-the-shelf
COTS
conforming to the manufacturer's datasheet and available to any purchaser

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Note 1 to entry: A single user is not able to influence the manufacturer's datasheet.

Note 2 to entry: This note applies to the French language only.

3.1.3
emulation
process that produces a substitute item for the purpose of replacing another item while maintaining the same form, fit and function, and interface

Note 1 to entry: Microcircuit emulation can replicate, with state-of-the-art, items that emulate the original item.

3.1.4
end of production
EOP
date of discontinuance from manufacture

Note 1 to entry: This note applies to the French language only.

3.1.5
equivalent item
item which is functionally, parametrically and technically interchangeable

Note 1 to entry: This term is also known as 'form, fit, function and interface' F3I.

Note 2 to entry: See also 'substitutes' (10.4).

3.1.6
existing stock
sub item(s) from within the supply chain and inventory that can be allocated to the item without requalification

3.1.7
indenture level
level of sub-division within a system hierarchy

EXAMPLE System, subsystem, assembly, and component.

Note 1 to entry: From the maintenance perspective, the indenture level depends upon various factors, including the complexity of the item's construction, the accessibility of sub items, skill level of maintenance personnel, test equipment facilities, and safety considerations.

[SOURCE: IEC 60050-192:2015 [2], 192-01-05]

3.1.8
item
subject being considered

Note 1 to entry: The item may be an individual part, component, device, functional unit, equipment, subsystem, or system.

Note 2 to entry: The item may consist of hardware, software, people or any combination thereof.

Note 3 to entry: The item is often comprised of elements that may each be individually considered. See sub item (3.1.21) and indenture level (3.1.7).

Note 4 to entry: IEC 60050-191:1990 (now withdrawn; replaced by IEC 60050-192:2015) identified the term "entity" as an English synonym, which is not true for all applications.

Note 5 to entry: The definition for item in IEC 60050-191:1990 (now withdrawn; replaced by IEC 60050-192:2015) is a description rather than a definition. This new definition provides meaningful substitution throughout this document. The words of the former definition form new note 1.

[SOURCE: IEC 60050-192:2015 [2], 192-01-01]

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3.1.9
integrator
organization that assembles an item of complex sub items for the end user

Note 1 to entry: Examples of items for the end user are: car, ship, aeroplane, oilrig.

3.1.10
life of need buy
LNB
procurement of sub items sufficient to support the item throughout its life cycle, or until the next planned upgrade

Note 1 to entry: Identified as a proactive LNB or a reactive LNB depending on which obsolescence approach is being used for the phase of the life cycle for which the procurement is taking place. (See 10.3).

Note 2 to entry: This term was previously known as lifetime buy.

Note 3 to entry: This note applies to the French language only.

3.1.11
manufacturer
organization or individual with responsibility for the design, manufacture, packaging and labelling of an item before placing the item on the market under their own name or trademark

3.1.12
obsolescence
transition of an item from available to unavailable from the manufacturer in accordance with the original specification

Note 1 to entry: In high reliability items, the sub item's original manufacturer and original specification are generally identified in the item's original configuration.

Note 2 to entry: Obsolescence might be because production has ended or because of the lack of availability of service provision, support of software or processed material.

Note 3 to entry: A specification can be of any form such as a drawing, datasheet, standard, written specification or a list of keywords and properties. A specification cancelled or an item's approval to a withdrawn specification can also cause obsolescence.

3.1.13
obsolescence issue
effect when the item is obsolete or when there is certainty as to when the item will become obsolete

Note 1 to entry: The item has an obsolescence issue when the PDN (see 3.1.18) has been issued by the manufacturer of the item.

3.1.14
obsolescence risk
measure of uncertainty as to when an item will become obsolete

Note 1 to entry: Obsolescence risk is often expressed in terms of a combination of the impact of an item becoming obsolete and the associated likelihood.

3.1.15
obsolete <tangible item>
no longer in production from the manufacturer in accordance with the original specification

Note 1 to entry: Examples include but are not limited to: materials, chemicals, components, electronics, and mechanical hardware.

Note 2 to entry: In high reliability items the sub item's original manufacturer, part number and/or original specification are generally identified in the item configuration.

Note 3 to entry: Discontinued is synonymous with obsolete.

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3.1.16
obsolete <intangible item>
no longer available from the manufacturer in accordance with the original specification

Note 1 to entry: Examples include but are not limited to: software, services, specifications, and processes.

Note 2 to entry: In high reliability items the sub item's original manufacturer and original specification are generally identified in the item configuration.

Note 3 to entry: Discontinued is synonymous with obsolete.

3.1.17
product change notice
PCN
notification from a manufacturer announcing a change of process, properties, characteristics or specification of an item

Note 1 to entry: This note applies to the French language only.

3.1.18
product discontinuance notice
PDN
notification of end of production of an item by the manufacturer

Note 1 to entry: This note applies to the French language only.

3.1.19
planned upgrades
predetermination points during the item life cycle at which the design of all or part of the item will be brought up to date by replacing sub items with better performing sub items and/or replacing obsolete sub items or sub items at risk from becoming obsolete

Note 1 to entry: These upgrades will often involve changes to both the software and the underlying hardware. Between the planned upgrades, one of the other obsolescence resolutions can be necessary (see Clause 10).

3.1.20
reclamation
use of a sub item that has a remaining useful life from a surplus or redundant item

Note 1 to entry: Reclamation is a controlled process (see 10.2 and recycled item in Annex A).

Note 2 to entry: Reclamation is sometimes referred to as cannibalization.

3.1.21
sub item
part of the subject being considered

Note 1 to entry: A sub item becomes the item, when individually considered.

[SOURCE: IEC 60050-192:2015 [2], 192-01-02]

3.1.22
supply chain
entire network of entities, such as organizations, people, technologies, activities, information and resources, directly or indirectly interlinked and interdependent, involved in serving the end user on delivering an item or service

Note 1 to entry: The supply chain is a part of an end-to-end process from the raw materials to the finished item delivered to the end user.

3.2 Abbreviated terms

COTS commercial off-the-shelf

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ECMP	electronic components management plan
EOP	end of production
IIOM	International Institute of Obsolescence Management [3]
IPR	intellectual property rights
LNB	life of need buy
OCM	original component manufacturer
OEM	original equipment manufacturer
OMP	obsolescence management plan
PCN	product change notice
PDN	product discontinuance notice
REACH	registration, evaluation, authorization and restriction of chemicals
RoHS	restriction of hazardous substances

4 Obsolescence management

4.1 What is obsolescence

Obsolescence is the transition of an item from available to unavailable from the manufacturer in accordance with the original specification.

Obsolescence is a significant cost driver and can occur at all phases of an item's life cycle. Unforeseen obsolescence issues can happen quickly, can cost a significant amount of unplanned resources to resolve and may put at risk an organization's financial stability, operations and reputation.

Obsolescence management activities ensure an item and its sub items can continue to fulfil their requirements over their expected useful life. Most items will become obsolete eventually; however, there is often uncertainty about when this will occur and the consequences to the item concerned.

Items include but are not limited to:

- physical assets (for example products, equipment, sub-assemblies, components, manufacturing tooling, test equipment, spares);
- consumables (for example ink cartridges, adhesives, computer media, batteries);
- materials (for example metals, fluids, polymers, composites);
- software (for example operating environment, middleware, firmware, specific applications);
- systems (for example information systems, classification systems);
- services (for example maintenance, communication and information).

For the purposes of obsolescence management, items may also include information and knowledge such as data, services, processes, procedures, specifications, standards and regulations.

Environmental legislation may have the potential to affect the use of some materials during the life of the item; those materials affected by the legislation should be considered from the outset of development.

Reasons why items become obsolete are:

- market factors including reduced demand of an item, lack of profitability or the introduction of new technologies and capabilities causing the manufacturer to issue a PDN;

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- changes to the configured item (for example configured item number, item updates, item rebranding, merger or acquisition of the manufacturer);
- obsolete sub items (for example raw materials or other sub items that make up an item);
- loss or change to manufacturing capability (for example process re-tooling, test equipment, manufacturing equipment, in-process materials, support equipment, documentation);
- loss of skill or knowledge to manufacture or support the item;
- loss of legal authority for manufacturers to place the item on the market due to regulations (for example export control), legislation (for example environmental protection legislation such as RoHS, REACH), or legal agreements;
- natural or man-made disasters.

NOTE See guidance on the effects of obsolescence on different technologies in informative Annex C.

4.2 What is obsolescence management

Obsolescence is inevitable but normally involves uncertainty as to when it will occur and what the likely consequences in terms of cost and impact will be. Due to this uncertainty, it is essential that obsolescence be addressed in a risk management context by understanding the level of risk and implementing any necessary activities to treat the risks. This makes it necessary to implement a management process to mitigate the negative effects of obsolescence.

Obsolescence management collectively describes the activities taken by individuals or an organization to manage the obsolescence of items. The objective of obsolescence management is to ensure that obsolescence is managed as an integral part of the item's life cycle: concept, development (design), realization (production), utilization (in-service support), enhancement (return to design upgrade) and retirement and then return to concept.

NOTE After the subclauses introducing the obsolescence management process and activities, the relationship between obsolescence management activities and the item's life cycle is shown in Figure 3.

Obsolescence management seeks to reduce the risk of obsolescence through undertaking planned activities to reduce the frequency of obsolescence issues and/or reduce the impact when an item becomes obsolete. The impact of obsolescence can vary depending on the type of item and the context in which it delivers value to the organization. Obsolescence impacts can be categorized in the following ways:

- Financial impacts. Responding to obsolescence can introduce significant costs to an organization from changing supply chains, design or testing new items, changes to item information or processes, or increased item cost as supply becomes scarce.
- Operational impacts. When an item becomes unavailable, the user can no longer derive benefit from the item. The unavailability of the item may also impact the user's ability to derive benefit from other sub items or items.
- Compliance impacts. The obsolescence of an item may prevent an organization from complying with regulatory requirements such as environmental or safety obligations.
- Reputation impacts. Where others are dependent on an organization to provide an item or service, the inability to provide the item or service due to obsolescence may have long term impacts on the reputation of the organization.

4.3 Benefits of obsolescence management

Effective obsolescence management typically delivers:

- an understanding of the current and future obsolescence risk being carried by an organization;
- a lower overall risk to an organization's objectives from obsolescence;

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- greater information and a framework to support obsolescence risk assessment and decision making;
- efficiencies through common obsolescence approaches, standards, processes or knowledge across and between organizations;
- more resolutions and time to respond to obsolescence risk and issues.

4.4 Obsolescence management process

The obsolescence management process contains the following steps, which are detailed in the clauses below:

- establishing an obsolescence management policy (see Clause 5);
- establishing an infrastructure and an organization (see Clause 6);
- developing an OMP (see Clause 7);
- developing strategies to minimize obsolescence during design (see Clause 8);
- determining an obsolescence management approach (see Clause 9);
- selecting obsolescence resolution and implementation (see Clause 10);
- measuring and improving the performance of the outcomes of the obsolescence management activities (see Clause 11).

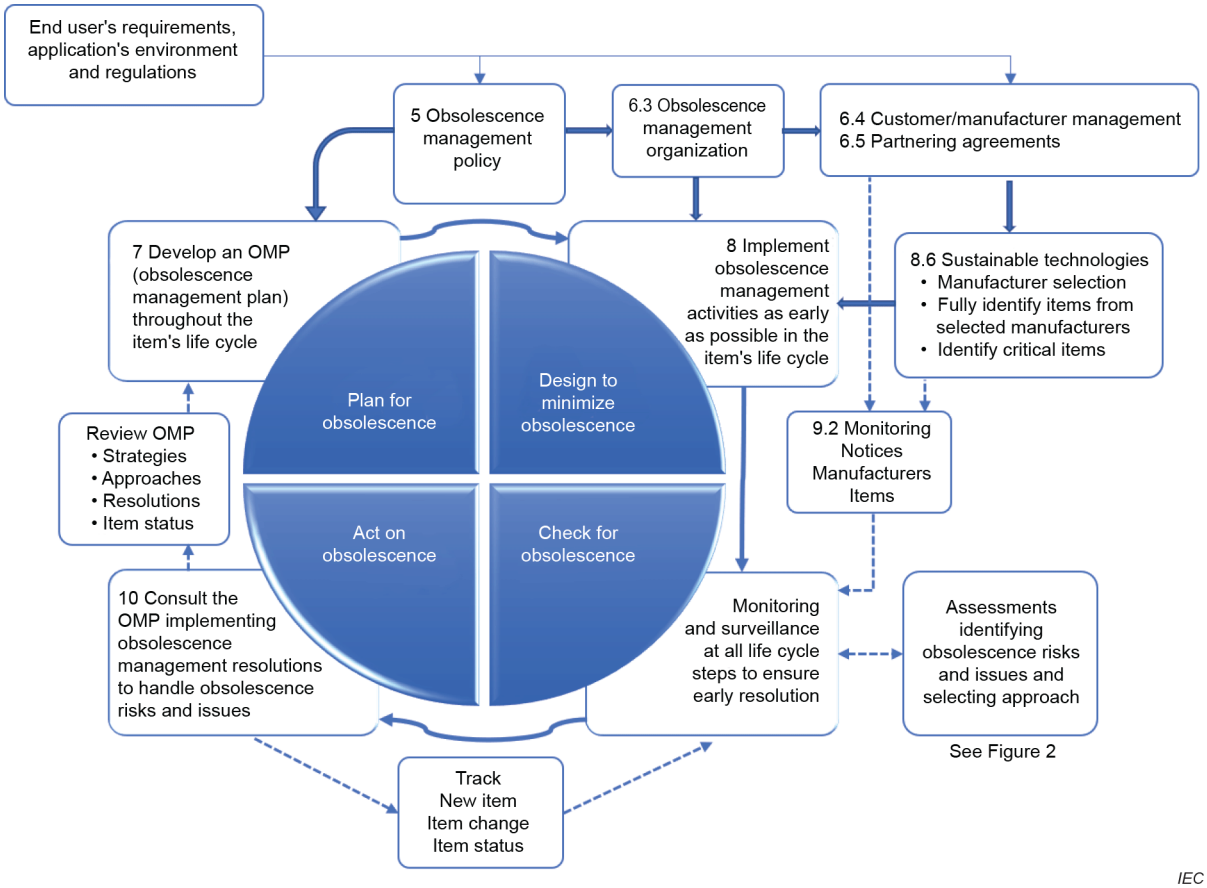


Figure 1 – Obsolescence management activities

Figure 1 describes the obsolescence management activities that should be considered throughout the life cycle phases of an item. The diagram has been simplified to identify the activities of an obsolescence management process.

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The end user's requirements (including technical, business and social) for the item have to be taken into consideration when performing and initiating the obsolescence management activities by setting an obsolescence management policy. The output of the obsolescence management process is a continuously evolving OMP. The obsolescence management organization sets up the necessary resources to implement the obsolescence management activities and strategies.

As each sub item is considered for the design it is selected from sustainable technologies and assessed, identifying obsolescence risks and issues for the life of the item. As part of the sustainment strategy, each sub item is fully identified (see 8.10) from the assessed manufacturers which are selected from the technology strategies (see 8.6).

Figure 2 details the obsolescence risk and issues box within Figure 1.

The general approach selected to treat the level of risk for each of the technologies of the item is added to the OMP, including approaches for critical sub items monitoring notices as necessary. All individual sub items may be recorded within the OMP or a reference document, with their selected approach, resolution and implementation, as necessary. During the life cycle of the item the status information of each sub item may change the level of risk, which may change the selected approach and resolution. This would result in the need to update with the changes during the monitoring and surveillance stage of the process.

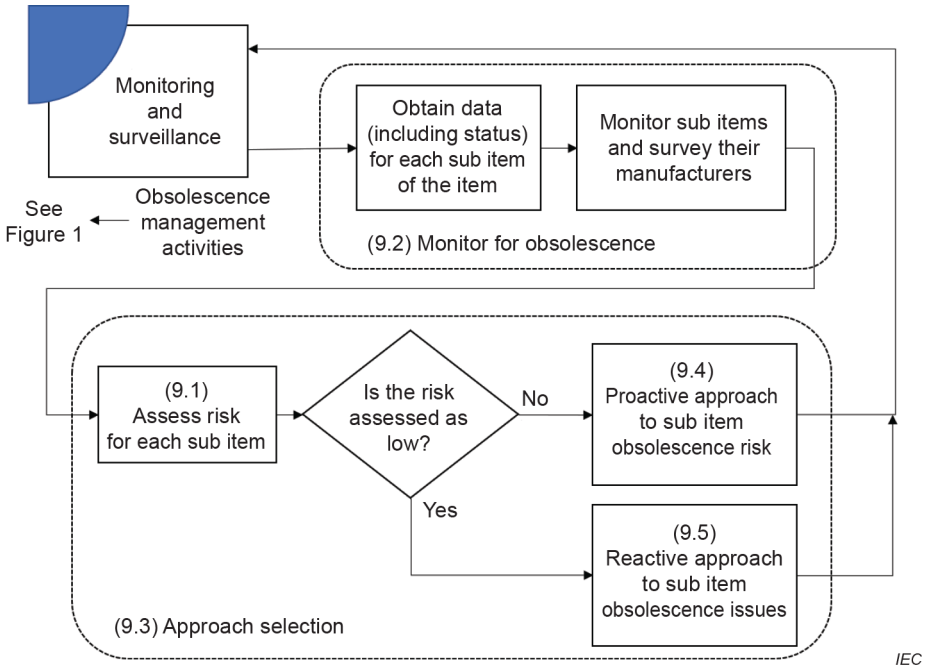


Figure 2 – Assessments identifying obsolescence risks and issues

The plan owner monitors the status of each sub item from their manufacturer's surveillance. The OMP should identify the level of risk for the technology of the sub item becoming obsolete which should determine the monitoring review period and therefore which sub items need monitoring regularly.

A list of examples of resolutions for each approach is given below. These have been suggested under either a proactive or reactive approach, but some can be interchangeable with either approach to treat the risk level.

A proactive approach to item obsolescence risks may initiate:

- design considerations (see 8.1):

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  - modularity (see 8.4);
  - technology transparency (see 8.5);
- planned design changes/upgrades (see 10.6);
- extended production (see 10.2, 2nd bullet);
- authorized continuing manufacturer;
- proactive life of need buy (LNB) to avoid obsolescence and materiel shortages (see 10.3);
- preservation – technology long term storage (for example die banking) (see 10.2 7th bullet).

A reactive approach to item obsolescence issues may entail:

- searching for existing stock (see 10.2, 1st bullet);
- processing an LNB reacting to a notification of obsolescence (see 10.3);
- substituting items (see 10.4) as an:
  - equivalent item;
  - alternative item;
- researching an aftermarket solution provider who may perform:
  - divestment – IPR transfer (see 10.2, 4th bullet);
  - reclamation (see 10.2, 6th bullet);
  - emulation/reverse engineering (see 10.5);
  - repair (see 10.2, 5th bullet);
- design changes/upgrades (see 10.6 minor design change);
- introducing a new item (see 10.6 major design change);
- no action – no impact (see 10.2 last bullet).

Figure 3 suggests periods when obsolescence activities may be used in an item's life cycle.

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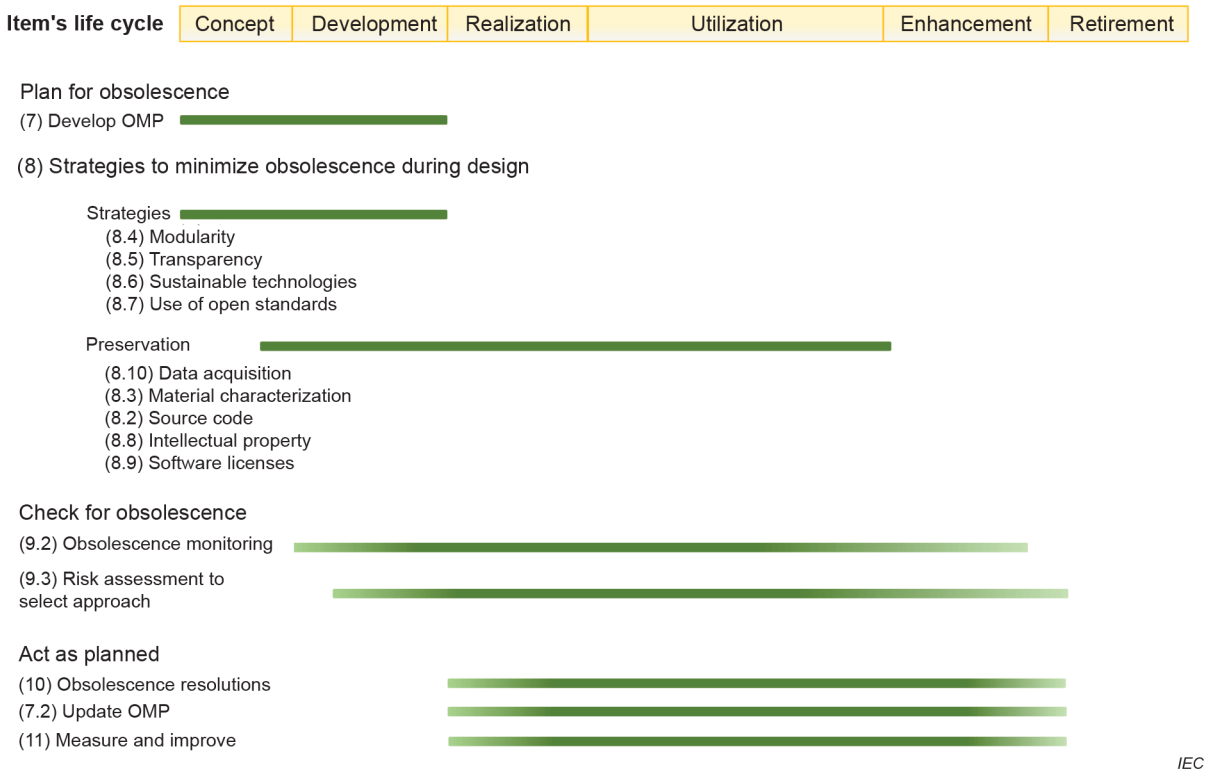


Figure 3 – Item's life cycle versus obsolescence management activities

5 Obsolescence management policy

An overarching obsolescence management policy should be in place to guide the overall effort to manage obsolescence. This policy may include integration of obsolescence considerations into an existing supply chain and supportability policy. Management shall enforce the obsolescence management policy where it exists. The policy shall identify recommendations of this document as requirements where required by an end user.

Usually only one obsolescence management policy is required for the whole organization, but the policy may allow for multiple OMPs.

The obsolescence management policy shall identify:

- the roles, responsibilities and infrastructure from all disciplines of the organization for managing obsolescence;
- the obsolescence management skill level and training required for the infrastructure;
- the organization’s operational procedures related to obsolescence management which may be part of a life cycle management plan or support plan.

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6 Infrastructure and organization

6.1 General

To execute the obsolescence management policy, there should be an obsolescence management infrastructure and organization in place. This infrastructure and organization should include:

- management responsibilities;
- obsolescence management organization and authorities;
- end user/manufacturer management;
- partnering agreements between organizations.

6.2 Management responsibilities

Top management should appoint a representative that is given the authority and resources to manage, monitor, evaluate and coordinate obsolescence management activities. The representative should be responsible for the obsolescence management policy and prepare and revise OMPs at all life cycle phases, approving revisions. This appointment is to promote effective and efficient operation and improvement of obsolescence management. The obsolescence management representative should report to top management and may communicate with end users/customers, contract manufacturers, integrators, and manufacturers on matters pertaining to obsolescence.

6.3 Obsolescence management organization

Obsolescence management has roles that should be performed to execute the obsolescence management policy which is suitably resourced according to the requirements of the organization. An individual can fulfill multiple roles and can be a dedicated member or a matrixed member of the obsolescence management organization. These roles should include but are not limited to: an obsolescence management lead communicating and/or coordinating with management, end users/customers, manufacturers and their suppliers, sourcing/contracting officers, engineering, quality, and finance. The organization should put in place the means for effective communication between all levels of the supply chain on identifying, treating and resolving obsolescence.

To enable staff to be able to identify, analyse and manage obsolescence effectively, they need to understand the processes that are involved in obsolescence management and therefore require the correct training and/or mentoring to enable them to perform.

The transfer of skills either through the physical transfer of resources or through training, may be necessary, removing reliance on a third-party organization.

NOTE Competence recognition for obsolescence practitioners is available from the International Institution of Obsolescence Management [3].

6.4 Customer/manufacturer management

The customer should check if there is capability to manage obsolescence. The customer/manufacturer relationship should allow for the timely exchange of information about technologies and items made obsolete, in order to facilitate the obsolescence management processes. Organizations should use this information and monitor the availability of their supplied items at regular intervals.

This relationship may be in the form of a contract between the manufacturer and their suppliers, ensuring early and comprehensive notification of errata, item status (PDN see 3.1.18) and change notification (PCN see 3.1.17) and an appropriate reaction to obsolescence issues from obsolescence monitoring (see 9.2.2). This contract may include

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other conditions such as traceability and warranties, and this relationship is often known as a franchised distributor for the supply of items from the manufacturer.

6.5 Partnering agreements between organizations

If applicable, contractual arrangements between manufacturers and/or integrators can be used to ensure that obsolescence information is actively communicated up the supply chain by the manufacturer, in order to provide an appropriate and timely response by the manufacturer and/or eventual end user. This includes any organizations within the supply chain where distributors have franchised agreements with their sources to provide items, including warranties. Items from organizations will be known as genuine when they have traceability back to the manufacturers, thus avoiding counterfeit items.

All organizations in the supply chain should have agreements which should include information exchange on obsolescence, appropriate time-limits and exchange formats. Information exchange between organizations within the agreement or contract should include the reporting of change and discontinuation notices (PCNs and PDNs) up the supply chain. It might not be to the manufacturer but to the integrator or the intermediate supplier/distributor depending on where in the supply chain or how complex it is in a particular technology/industry.

NOTE For further information on exchange formats see VDMA 24903 [4].

As part of any partnering agreement between organizations, the assessment of capability in relation to obsolescence management may be included in the quality management system for approved manufacturers and their approved suppliers. These partnering agreements may be in the form of a service level agreement, contractually committing a supplier to the level of item availability, thus devolving the risk of obsolescence onto the supplier.

7 Development of an OMP

7.1 OMP

In accordance with the obsolescence management policy, an OMP shall be developed and implemented to ensure appropriate selection and timely implementation of relevant obsolescence activities. The plan should include strategies to minimize obsolescence during design as well as reactive or proactive obsolescence management approaches for technologies, as appropriate, to the risks identified.

The OMP describes the activities for prevention, detection/identification and treatment of the effects of obsolescence through all phases of the item life cycle, to achieve the optimum compromise between life cycle costs for the item's performance and item availability, maintainability and safety. The OMP can be part of another plan or a separate document.

7.2 OMP contents

The OMP contents include but are not limited to the following:

- Authorities and resources. The plan shall identify the organization or individual for developing, reviewing and maintaining the plan and should identify the milestones for future transfer of ownership of the plan, if applicable. The plan shall identity who is the organization or individual to approve the plan. The plan shall identify the skill level and roles to execute the obsolescence management activities of the plan and should identify financial resources.
- Scope. The plan shall identify:
  - the items being considered for obsolescence management;
  - the length of each item’s life cycle;
  - the organization’s obsolescence management policy (should list the contents);

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  - the customer/end user' requirements (should list the references).
- Objectives. The plan should identify the objectives of the obsolescence management activities which may include:
  - data acquisition;
  - strategies to minimize obsolescence during design;
  - item monitoring;
  - item risk assessment;
  - approach selection;
  - resolution selection and implementation;
  - review of the plan;
  - measurement and improvement.
- Data acquisition. The plan shall identify the data required to determine the obsolescence management approach and make it more efficient.

  NOTE 1 Data includes quantitative and qualitative information. See 8.10 for guidance.
- Strategies to minimize obsolescence during design. The plan can identify strategies throughout the item's life cycle which include:
  - surveillance of sustainable technologies and their assessed manufacturers;
  - selection of sub items from sustainable technologies to minimizes obsolescence;

    NOTE 2 Selection of sub items from assessed manufactures can reduce the risk of obsolescence for the life of the item.
  - plan item upgrades for unsustainable technologies.
- Flow down of obsolescence management requirements throughout the supply chain. These should include agreements for timely receipt of notices of obsolescence (see also 6.4 and 6.5).
- Monitoring and schedule. After the initial surveillance of technologies, selection of manufacturers and monitoring of item and sub item status, the plan should schedule those items that require monitoring. Those sub items that are identified as critical to the item shall be scheduled for periodic monitoring, relevant for that technology.
- Obsolescence management approach. The plan should define the obsolescence management approach to be implemented for each sub item of the item.
- Resolution selection and implementation. The plan should implement the most cost effective resolution to treat or resolve obsolescence.

  NOTE 3 This could be referenced to by leveraging another document with the process already described.
- Review of the plan. The plan should include the obsolescence reviews to be undertaken throughout the life cycle to an agreed schedule and criteria. This will determine the suitability, adequacy and effectiveness of the obsolescence management activities including the obsolescence resolutions.

  NOTE 4 Reviews involve investigating the current situation to compare it with what is expected or required, in order to determine discrepancies which can be analysed so that improvements can be recommended.
- Plan performance. The plan should document the metrics and measurement for the process improvement.
- Tools used. The plan shall document details of how relevant information should be documented and stored and for how long it should be kept.

  NOTE 5 For further guidance on what can be included in an OMP see informative Annex D.

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8 Strategies to minimize obsolescence during design

8.1 Obsolescence as a consideration in design

Where possible the risks arising from obsolescence should be reduced during design, whenever a design activity is undertaken. The choice and the selection of items (for example materials, components and interfaces) should be made in order to minimize the risk of obsolescence.

Consideration should be given to:

- changes to regulations which may affect the market;
- investigating multiple sourcing;
- monitoring obsolescence (PDNs see 3.1.18);
- monitoring changes (PCNs see 3.1.17).

For example, selection of an item with a long-term production life, which is available from several manufacturers and compliant with regulations, reduces obsolescence risks.

Electronic component management plans governing the selection of hardware typically address obsolescence management (see for example IEC 62239-1 [5]). A similar approach should be considered for other technologies.

The following subclauses 8.2 to 8.10 are strategies that are best considered during the development phase to control risks and costs arising from obsolescence; however, the strategies could still be implemented in later phases of the item's life cycle.

8.2 Source code

All the source code, licences and associated documentation for software modules as well as tooling and skills needed for software development/production should be preserved to ensure appropriate maintainability and future enhancements.

8.3 Material characterization

Raw and manufactured materials should be characterized so that they can be comprehensively specified to an alternative manufacturer if this becomes necessary. There may be a need to assess an alternate manufacturer while the item is still in production.

Material characterization is mainly relevant for use with:

- raw materials;
- manufactured materials;
- manufacturing processes.

8.4 Modularity

Modularity is the use of exchangeable sub items (for example parts, products, sub items, and software) and should be implemented to make the item easy to replace and repair.

8.5 Transparency

Transparency is a design methodology that is based on the specification of interfaces. Transparency should be implemented to allow any technology to be used in manufacture and support provided that the form, fit and function and interface F3I of the individual items (for example component or module) is maintained.

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8.6 Sustainable technologies including materials

The process for selecting sub items should consider the following in order to make technologies and items less likely to become obsolete:

- technologies and items that are likely to have long life cycles and where the ability to forecast future changes exists;
- assessed manufacturers for each technology that meets all the requirements of the design and regulations where procurement legislation allows this approach;
- manufacturers based on research into market trends and where possible, having multiple assessed manufacturers listed against the items within the technology.

8.7 Open standards

It is important to develop the architecture/design of the item to facilitate the replacement/repair of obsolete sub items (hardware and software), so that readily available substitute sub items (see 10.4) can be used in place of obsolete sub items. Implement open standards, protocols, and interfaces to allow for easier modular approach for technology upgrades. An open architecture reduces obsolescence costs, because it avoids expensive design changes by facilitating the insertion of advanced technologies and software. Often, it also enables competition, which will minimize the likelihood of future obsolescence issues. The architecture of an item should be designed to take growth and evolving standards and interfaces into consideration. This provides for change, while minimizing the impact on existing item functions.

In addition, if applicable, the design should allow for partitioning of the software into appropriate sub items that can be tested in isolation and should avoid making software dependent on the hardware through appropriate isolation of drivers and the use of abstraction layers.

8.8 Obtaining IPR

IPR can restrict the legal rights of the organization to change or reproduce item designs without reference to, and contracts with, the owner of the IPR. Where possible, intellectual property should be obtained for items determined to be at high risk when it is cost effective.

8.9 Software licensing

Licences for software sub items should be available during the complete life cycle of the item. Specific contracts may be needed with software manufacturers to allow the usage of software when the software is no longer supported.

8.10 Data acquisition

To manage obsolescence efficiently, availability of data such as the status of sub items and item indenture level should be shared with everybody involved (for example design office, supply chain, procurement). Data shall be acquired to ensure that the likelihood and impact of obsolescence risks are evaluated. This data can include but is not limited to:

- expected market life of the item/sub items;
- a list of configuration sub items within an item;
- the identification of the items and sub items details: manufacturer, part number and specification. Ensure the item details correctly identify the manufacturer of the item/sub item and not an agent or catalogue supplier’s name and not a stock number. For physical items ensure the specification or datasheet identifies the item’s part number and marking details;
- failure rate and expected useful life of items;
- demand rate/spares consumption rate;

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- trend of regulations, manufacturing technologies, software change intervals, etc.;
- technology plans;
- life cycle data (for example EOP forecast of sub items);
- notices (PDNs, PCNs, errata);
- custom built data/intellectual property;
- planned upgrades including implementation costs;
- inventory levels/spares information. Obsolescence management includes the utilization phase of the item's life cycle. To assist support, ensure spares procurement is initiated sufficiently in advance to maintain the minimum spares level, which itself requires a level of management to assess the spares demands and levels;
- storage life of items, cost of storage and level of degradation whilst in storage;
- cost information to support cost benefit analysis.

NOTE 1 Access to sub item data can also be obtained from third party obsolescence management tools in the market place to assist in identifying the obsolescence status of items. It is important to choose a tool that is suitable for the application, as these tools can differ in the services that they provide and do not cover all types of items.

NOTE 2 Standardization guidance on an items database with a data dictionary is listed in the Bibliography [6] [7].

9 Obsolescence management approach

9.1 Introduction to risk assessment

In the context of obsolescence management, risk assessment provides decision-makers with guidance on the approach to be taken to manage obsolescence based on an understanding of when obsolescence is expected to occur and its likely impacts. A level of risk can be estimated from these parameters and, in general, this estimate will determine whether a reactive or proactive approach should be taken. It will also inform how much effort should be put into treating the risk, as the cost of managing the obsolescence of an item should be commensurate with the risks associated with its obsolescence. The level of risk should be determined based on a number of factors including a trend of demand rate, technological change, social requirements and impact of regulations or legislation (see 4.1).

An understanding of when impacts are likely and their magnitude can also inform, in broad terms, how obsolescence risks should be treated, for example whether to target impact reduction, likelihood reduction or reduction of uncertainty.

For high risk items a more rigorous analysis of risk can be useful to further understand causes and consequences so as to select the most appropriate resolution.

Principles of effective risk management are set out in ISO 31000:2018, Clause 4 [8], and good practice in risk assessment is described in IEC/ISO 31010 [9].

The risk of obsolescence needs to be assessed at all phases through the item life cycle as early as possible in the life cycle, from concept and development phases where the item is selected. However, only limited information may be available. As the information matures, the analysis should be progressively developed and reviewed.

Undertake the risk assessment process periodically during the realization (production) and utilization (operation) phases to avoid, for example, item delivery shortages or item maintenance issues.

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An initial filter may be applied where some items can be immediately determined as having a low obsolescence impact and therefore a reactive approach is appropriate for those items. (For example, fasteners designed to a standard such as nuts, screws or bolts would not need a further risk assessment).

The results of the risk assessment select a proactive or reactive obsolescence management approach, or more likely a combination of the two. Figure 4 shows a summary of the resolutions in relation to the sub item life cycle and the approaches selected.

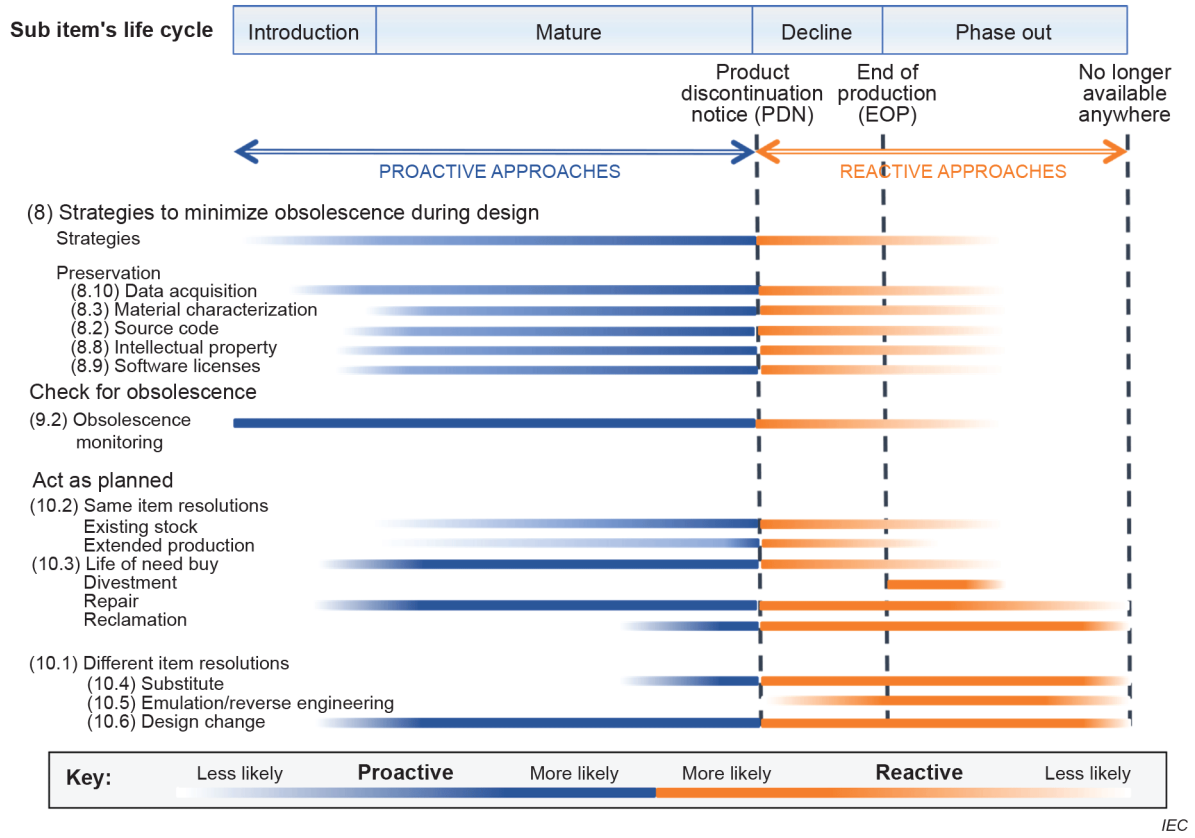


Figure 4 – Proactive versus reactive approaches with resolutions

9.2 Obsolescence monitoring

9.2.1 Monitoring background

Obsolescence monitoring involves tracking items and their manufacturers, to determine the item's current availability status and, where possible, when the item will become obsolete.

NOTE Some specialized tools can support the search of the item availability status.

To be effective, monitoring requires accurate data. The sub item list supplied with a new item should make it possible for ongoing detailed monitoring of obsolescence. In the case of a legacy item without detailed records, it may be appropriate to carry out an obsolescence survey to scope the extent of likely problems such as unavailability of COTS and standard items. Monitoring should be conducted at a frequency that meets the needs of the organization and corresponds to the risk that was evaluated in the risk assessment. It should provide assurance that obsolescence can be identified with sufficient time to allow for selection of the optimal resolution. When online services provide the information, monitoring can be in real time. More manual methods should be evaluated to determine the optimum frequency.

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Surveillance of all the data referred to in 8.10 from the manufacturers should be monitored including: notices (closure, changes/PCNs or obsolescence/PDNs) and changes to names of organizations and their agreements to establish if there is an effect on the obsolescence risk to the item.

9.2.2 Obsolescence notice monitoring

One form of monitoring is based on notices of obsolescence. These notices can be in many forms depending on the technology and the organization's processes. Careful review of the manufacturers' notification documentation, including 'errata', may be necessary. The title of the notification does not always reflect its meaning and may require obsolescence activities.

NOTE 1 The electronics industry normally issues a PCN or a PDN from the manufacturer to indicate the items that are to change or the production of which will cease with a last buy date.

NOTE 2 The software industry and electronic equipment manufacturers are likely to be affected by vulnerability notices, updates, item errata and new features adding onto existing software.

The agency responsible for managing the technical aspects of the legislation should be monitored to identify any change of status of hazardous substances.

NOTE 3 The materials industry issues bulletins and notices for regulations where a material containing a hazardous substance is subject to increased obsolescence risk if the substance becomes identified for more stringent legislative restriction on use. See IEC 62474 [10] for the declaration of substances.

NOTE 4 There are commercial organizations that aggregate PCNs, PDNs, and other data on items. This information can be very useful in providing the current production status of these items. See Bibliography for further guidance.

9.2.3 Direct contact monitoring

If no information systems are available to facilitate monitoring and surveillance, as is the case with many COTS items and non-electronic items, monitoring can be achieved by using direct contact with the manufacturers, communicating using standard methods.

NOTE Sometimes direct contact with the manufacturers is not possible due to low volume of procurement; in this case the contact is generally made through the franchised distributor.

The customer/manufacturer relationship should be leveraged to allow for the exchange of timely information about obsolescence. Organizations should use this information and monitor the availability of their supplied items at regular intervals. If applicable, contractual arrangements between organizations should be used to ensure that obsolescence information is actively moved up the supply chain by the manufacturer, in order to provide an appropriate and timely response by all the affected levels in the supply chain and/or the eventual end user.

Sub item monitoring entails research to best determine both when the sub item will become obsolete and the time until the item will be impacted. Some surveillance can be done via research directly on the manufacturer websites while other manufacturers need to be contacted. The data that should be collected includes planned item upgrades (obtain information on new features, versions, add-ons, functionality), updated item support information (vulnerability updates, end of support), assessment of manufacturer stability, and upcoming releases and related items.

Market analysis and monitoring should be completed by reviewing all hardware and software items together. This analysis should consider the criticality of the software and impact on the item, the number of years required for sustainment, and the current number of forecasted years until the item will be impacted.

9.3 Risk assessment to select approach

The main steps for assessing obsolescence risk are as follows:

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a) develop a list of all items where the potential for obsolescence impacts can be determined;

b) establish criteria to be used for risk assessment;

c) obtain the necessary data to perform the risk assessment;

d) assess and quantify the likelihood and impact of obsolescence.

Undertake the risk assessment process, at the relevant indenture level for the item, to derive the level of risk to determine if a proactive or reactive approach is required for each sub item. A reactive approach should be selected where the risk significance is low. In all other cases, a level of proactivity should be selected, aiming to reduce the likelihood of an occurrence of obsolescence and/or decrease the impact when obsolescence occurs.

Obsolescence involves uncertainty as to when it will impact, as such the likelihood is characterized in terms of time (rather than a probability as used in risk assessment within ISO 31000 [8]). Examples of likelihood of obsolescence include “the probability of the item being obsolete today” or “the point in time when the item is forecast to become obsolete” or “probability that an item will become obsolete in a specified time period”. The time period for the likelihood assessment should be defined in accordance with the period in which proactive activities can be planned.

NOTE 1 See Annex E for examples of obsolescence risk assessments.

The following should be considered when assessing obsolescence risk over the life of the item:

- the likelihood of an item becoming obsolete (for examples see 4.1). Forecasting obsolescence, technology and market trends needs understanding of when a technology or an item will become obsolete. Forecasting will allow the obsolescence impact of an item to be reduced;
- factors that delay the impact of an item being obsolete, for example current stock level or demand for the items needed to produce or maintain the capability;
- impact of an item being obsolete, for example:
  - criticality of the sub item to the function of the item;
  - time it will take to resolve an obsolescence issue considering factors such as the availability of technical specifications, robustness of the industrial sector that would be involved in the resolution, the use of materials, the technology needed to manufacture the item, testing requirements;
  - impact of software obsolescence and the supporting infrastructure - the effect on information assurance and operating environment compatibility.

NOTE 2 Predicting changes to environmental regulations can have an impact on materials within items and cause them to become obsolete. Understanding if this has an impact on an item can be difficult especially if the item is being manufactured at a low level within the supply chain.

Software, for instance operating systems, middleware, application software, drivers, and firmware, may or may not have a major impact on the item depending on how dependent the item is on certain hardware and/or software sub items. The risk assessment needs to consider the potential impact on both hardware and software should one or the other go obsolete. This impact may not just affect the availability of the item but also the security of using the item too.

All interfaces and software source code should be fully documented and archived such that the consequences of obsolescence can be determined. It is recommended that additional documentation to cover the internal application code and modules’ use be prepared.

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9.4 Proactive approach

When a proactive approach has been selected actions need to be taken prior to an item becoming obsolete and prior to an actual obsolescence impact. This should be proportionate to the level of risk. The proactive approach treats the risk of an obsolescence impact and is a best practice in a high majority of cases. (The opposite reactive approach does not treat the risk).

NOTE If the item is expected to be replaced with an equivalent item and requires no integration, then it can be adequate to take a proactive approach and monitor the item and identify an appropriate resolution if obsolescence occurs. Whereas for a COTS item manufactured from only one manufacturer and requiring a high level of integration, an LNB (see 10.3) or long term supply/support contract can be a more appropriate approach.

Factors such as item criticality and the likely timeframes or cost of potential resolutions will influence the decision to employ proactive approaches. For example, proactive approaches are often appropriate for high criticality items or items with short notice periods for LNB. See Annex C for guidance.

Planned upgrades (enhancement life cycle phase) are another element of a proactive approach. Design strategies to identify technologies of sub items with long production cycles compared with the item's expected life can limit the number of upgrades. Planned upgrades should be used at predetermined points during the item's life at which the design of all, or some sub items, of the item will be brought up to date and obsolete sub items replaced. The item upgrade programme should consider the need to minimize life cycle costs. Planned item upgrades do not necessarily preclude the need for proactive monitoring. Between the planned upgrades, at least one of the other resolutions for dealing with obsolescence will be needed. An LNB may be appropriate for example.

Proactive activities should be appropriately aligned with any future capability upgrades.

9.5 Reactive approach

A reactive approach is to do nothing until obsolescence becomes an issue.

This reactive approach should only be selected for items that are assessed as having a low impact, for example, items designed to a standard (for example military specification), items with multiple assessed manufacturers, items with low likelihood of obsolescence, for example mechanical items and machined parts, and items that will not fail in the life of the capability. A reactive approach may also be assigned where the risk of obsolescence impact is extremely low. Selecting a reactive approach in the initial phases of the life cycle will limit the resolutions that can be selected in the future.

10 Obsolescence resolutions

10.1 Resolution selection and implementation

"Resolutions" is the collective term for the different types of activities that either treat the risk of obsolescence impact or overcome an obsolescence issue. Resolutions should aim to minimize the overall impacts of obsolescence, and the selection and implementation of resolutions will depend on numerous factors including:

- the reasons for obsolescence;
- the feasibility and potential cost of resolutions;
- the second order impacts on supportability or future requirements for the item;
- potential timing of resolutions;
- permanence of the issue or resolution.

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A record of the factors analysed, the decisions taken, and the resources required in implementing the most appropriate cost-effective resolution should be made. The implementation phase of the resolution may follow the organization's configuration and control processes; these should be monitored.

When selecting resolutions, it is also important to consider what approval may be required. All resolutions are business decisions and will generally require some form of commercial approval. Additionally, depending on the type of resolution selected and the purpose or criticality of the item, technical approval may also be required, for example:

- design changes will often require approval from a design authority;
- substitutions may require item qualification.

Where a resolution involves a substitute or design change, strategies to minimise obsolescence during design (see Clause 8) should be considered.

Resolutions may involve one-off activities or occur multiple times in the life of an item. Multiple resolutions may also be selected to effectively manage obsolescence risks and issues. Common resolutions include:

- replacement with the same item;
- LNBs;
- substitutes;
- emulation and reverse engineering;
- design changes.

The following subclauses 10.2 to 10.6 will describe these common resolutions and their particular considerations. See informative Annex B for examples of resolutions with different scenarios. Also see informative Annex F for examples of an obsolescence management decision process for selecting resolutions.

10.2 Same item

Same item resolutions include the following:

- existing stock: the conduct of an item search to find an alternate source for the same item from stock that has been purchased from the manufacturer;
- extended production: this involves negotiating with the manufacturer to continue production of the item;
- LNB (see 10.3);
- divestment: this is where the manufacturer has sold the assets to another organization or passed on the intellectual property of a technology to an aftermarket manufacturer (see A.1);

  NOTE 1 Intellectual property escrow consists in transferring intellectual property (for example engineering designs) to a third party (an escrow agent) on behalf of the end user and the manufacturer and contracting the third party to disburse the intellectual property based on conditions agreed to by the transacting parties, for example transferring the intellectual property to the end user if the manufacturer goes into liquidation.

- repair or contract support: it may be possible to repair an obsolete item that cannot be replaced, according to the original specification. Care should be taken to ensure compliance with applicable rules and standards, particularly in sensitive fields such as aerospace or nuclear plants. Repair work needs necessary technical facilities, tools and trained operators to carry out professional and sustainable repairs. Electronic items in particular (but not exclusively) should be tested under realistic use conditions after repair. Repairs should be sustainable, strategically replacing items subject to wear. Repairs should be documented in a repair report;

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NOTE 2 For software, design changes or bug fixes can be accommodated by contracting for support from the manufacturer or from a franchised third party. When contract support for software is likely to be the chosen option, at the start or later in the life cycle of an item, the software acquisition contract can make provision for sufficient documentation, support facilities and IPR to be available so that franchised third party software maintenance is a realistic option.

NOTE 3 Restoration/repair. It is often possible to professionally repair an item even after it has been discontinued by the manufacturer. In some cases, the manufacturer can still render this service, even after discontinuation. In addition, it is also possible to carry out the repairs internally or by employing an industrial service specializing in repairs to obsolete items.

- reclamation: this is the process of re-using items from within the inventory that have a remaining useful life; each reclaimed item should be checked to see if re-qualification is necessary;

NOTE 4 Certain regulatory agencies do not permit this to happen for some items.

NOTE 5 This process describes the obsolescence management resolution of taking items and assemblies from old stock in order to reuse it in a different position. This resolution is associated with certain risks which are considered. As a basic principle, all reclaimed items are evaluated to determine reliability and the expected remaining service life. This includes a check to see if repairs, preventative measures or procurement of replacements are possible and remain possible for the expected duration of use.

NOTE 6 Reclamation or recycling, (see the definition of recycled component in A.13), as it is commonly known, is a major source of fraudulent recycled electronic items, see IEC 62668-1 [11].

- preservation: consider methods for preserving designs, tooling and sub items during the obsolescence support phase of the item's life cycle. For example: IEC 62435-1 [12] for the finished electronic component's long term storage. The preservation review should include identification, handling, packaging, storage and protection of the item. Consideration should be given to the selection of the most appropriate media to store documentation and software. Software stored on media should be subject to periodic re-verification to ensure that the contents remain accessible for the support period;
- no impact: in the case of software with perpetual licenses, the engineering team decides an outdated version of the software will not impact the item and no changes are required. If long term storage of software code is required, entering into an escrow arrangement with the manufacturer to store the source code may be necessary in case future modifications are needed.

10.3 Life of need buy

A life of need buy (LNB) is a same item resolution which involves purchasing the quantity of relevant hardware items or software license items predicted to be required for a defined period. It can cover the complete amount for a production run and associated spares or it may cover only the sub items known to be at risk during support activities. A decision to undertake an LNB should take account of planned upgrades to the item. It will often become appropriate when a manufacturer makes known an intention to cease manufacture of a particular item to undertake an LNB with no known suitable substitute or a software manufacturer ends sales of a particular version of a software item.

An LNB may be considered by any manufacturer in the supply chain, on its own, or in cooperation with other manufacturers of the same item. Use of an LNB avoids issues with IPR. Significant shortcomings to performing a LNB are the financial implications and the uncertainty of the quantities required.

When considering an LNB the risks associated with long term storage of the item along with risk such as the possibility of loss or damage should be considered in conjunction with the usage rate. The appropriate storage conditions would therefore have to be analysed to achieve successful storage, as some items may require special storage conditions. Some items may require periodic inspection, analysis and testing to ensure their suitability for use (refer to IEC 62435-4 [13] for the storage of sensitive semiconductor parts, including semiconductor die and wafer banking).

A LNB should be considered:

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- when there is a known, or forecast of, EOP;
- when the life expectancy or production of an item is short;
- when an item is procured to satisfy an urgent operational need;
- when difficulties caused by future design changes of the item need to be avoided;
- when there is low demand;
- when there is enough storage capacity;
- when items' shelf-life allow long term storage;
- in order to avoid difficulties caused by sub item design changes by its manufacturer, resulting in subtle changes in the construction of items.

In a continuous improvement approach to the OMP, such as measuring cost avoidance, it can be necessary to identify an LNB as a proactive need buy or as a reactive need buy depending on when the procurement took place during the life cycle and whether it was planned.

After an LNB is made, the usage of items should be tracked to determine whether the assumptions used to calculate the life-of-need quantity were correct. If the demand is greater than expected, it may be necessary to put another resolution in place.

10.4 Substitutes

Substitutes include the following:

- Equivalent items: These are items which are functionally, parametrically and technically interchangeable with the obsolete item. This includes minor software security updates, which are backwards compatible. Minor testing may be required to validate the use of an equivalent item.

  NOTE 1 Equivalent items are sometimes referred to as simple substitutes.

- Alternative items: These are items that provide a limited parametric match, but which, after consultation with the person or organization responsible for the design are an acceptable application specific alternative for the obsolete item. This includes major software updates or changes that may result in use of software provided by a different manufacturer, running on a different platform or operating environment, and providing similar, but not identical functionality. Major testing may be required to validate the use of an alternative item.

  NOTE 2 Alternative items are sometimes referred to as complex substitutes.

Challenges include ensuring that the substitute items are fully assessed and qualified to meet the end user’s item requirements. Once the substitute items are approved, the substitute item’s details, manufacturer, full part number and/or specification, should be added to the item’s configuration. Keeping track of manufacturers’ names and notices associated with these substitute items may be achieved by maintenance of a suitable database.

NOTE 3 There are online databases that can facilitate the search for substitute electronic items. Similar tools do not exist for all other types of items.

10.5 Emulation and reverse engineering

Emulation is a process that produces an equivalent item using either the original specification or from characteristics gained from an in-depth examination of a working example of the item to be replaced. Reverse engineering is a technique which is used when the original specification is unknown or incomplete. Reverse engineering allows development of a new technical specification that can be used to manufacture a new item. If the original specification is known, the item can have a design change to the same specification or different specification. This process can involve significant cost and time and can involve complex IPR issues.

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Software emulation of the operating environment and applications could be a way to resolve software obsolescence issues where the existing software can no longer run on the proposed hardware environment. In addition, one should consider changes in performance (from potentially using faster processors), stability, IPR, and support for the emulation software.

10.6 Design change

A new design is done to resolve obsolescence by means of updating or upgrading the item, as well as enabling the use of newer items.

A design change should be considered when it is not possible, practical, or cost effective to procure or emulate a substitute item. There are two levels of design change:

- minor design change (for example on the next level assembly);
- major design change (for example item replacement).

A design change can also be used to treat multiple lower level obsolescence issues when that approach is more cost effective. Any design changes should include an assessment of the need for re-qualification. Design change should be performed with various types of assessment, such as cost efficiency, reliability, item life cycle cost, maintainability, cost of stock and storage. Planned upgrades can also be considered at the start of programs which take the place of design changes and can replace both hardware and software.

11 Measurement and improvement of obsolescence management activities

11.1 General

A metrics section should be included as part of the OMP, which should identify what metrics or measures need to be collected, how the team will collect the results, when or how often the collections should take place, and who needs to review those measures and analyse the results. Clause 11.2 lists the metrics required to track the performance of the outcomes of the obsolescence management activities. Metrics help with analysing the value that obsolescence management is providing, both reactive and proactive. These metrics may be derived from higher level business metrics and may also identify areas of improvement within the process.

11.2 Metrics

Obsolescence management metrics focus on the performance of all the obsolescence management activities (for example the obsolescence management team, the effectiveness of the plan, and the processes). These metrics show whether the obsolescence management approach is meeting the objectives set out in the OMP. Metrics can also provide an indication of where changes can be made if objectives cannot be met. Examples of some possible metrics include:

- the effect of obsolescence on trends in overall sustainability, supportability, and availability of the item;
- the cost penalties resulting from service unavailability, because of unresolved failures due to obsolescence;
- the average cost of implementing the OMP (i.e. management activities + resolutions) over the life cycle of the item – assessed by resolution type;
- the percentage of the configured list that is obsolete;
- the percentage of proactive resolutions versus reactive resolutions;
- the accuracy of forecasting;

NOTE 1 Comparing the predicted EOP date with the actual date of issue of the PDN can be a measure for an internal process or a request for more information from a data provider.

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- the percentage of reactively managed items that should be proactively managed but are not due to limited resources (time, people, funding);
- the number of reactive obsolescence issues found for items that were proactively managed (defects);

  NOTE 2 These defects can be defects per month, defects per items monitored, or the impact of defects per item monitored. Defects can include issues such as false obsolescence notifications and reactive obsolescence found on items being proactively monitored.

- the number of active obsolescence issues currently open;
- the average time from opening to closing an obsolescence issue;
- the percentage of PDNs not being received and addressed within the appropriate timescale;
- the money spent on proactive and reactive obsolescence management per proactively managed item;

  NOTE 3 This is the impact of each incorrect decision where a reactive resolution was chosen instead of a proactive resolution. The impact can be measured in either hours: consequence of schedule, or cost: return on investment, productivity metrics, cost avoidance, down time, length to repair, spares availability, counterfeit issues.

- the end user requirements and stakeholder satisfaction met versus not met.

The metrics chosen should be suited to the business need or process maturity.

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Annex A
(informative)

Vocabulary relating to obsolescence

Annex A lists terms and definitions commonly used in electronics, aerospace and defence.

A.1
aftermarket manufacturer
organization that produces items through emulation, reverse-engineering, or design change, that match the manufacturer's specifications and satisfy customer/end user needs without violating the manufacturer's IPR

Note 1 to entry: Those organizations which do have a contract with the manufacturer to produce items are known as 'authorized manufacturers' or 'authorized aftermarket manufacturers' when the manufacturer has made the items obsolete. These items are produced from sub items that have been transferred from the manufacturer to the aftermarket manufacturer or produced by the aftermarket manufacturer using the manufacturer's tooling and/or intellectual property.

A.2
aftermarket solution provider
organization that specializes in processes to solve an obsolescence issue

EXAMPLE Aftermarket manufacturer, authorized aftermarket manufacturer, aftermarket source.

Note 1 to entry: Processes currently identified are: remanufacturing, die reclamation, adaptor/daughter boards, uprating, up-screening, re-balling.

A.3
aftermarket source
reseller which may or may not be under contract with the original component manufacturer (OCM) or is sometimes a component "re-manufacturer", under contract with the OCM

Note 1 to entry: The reseller accumulates inventories of encapsulated or non-encapsulated (wafer) components whose end of life date has been published by the OCM. These components are then resold at a profit to fill a need within the market for components that have become obsolete.

[SOURCE: IEC 62668-1:2019 [11], 3.1.1]

A.4
broker
individual or corporate organization that serves as an intermediary between buyer and seller

Note 1 to entry: In the electronic component sector a broker specifically seeks to supply obsolete or hard to find components in order to turn a profit. To do so it can accumulate an inventory of components considered to be of strategic value or can rely on inventories accumulated by others. The broker operates within a worldwide component exchange network.

[SOURCE: IEC 62668-1:2019 [11], 3.1.2]

A.5
counterfeit materiel
materiel whose origin, age, composition, configuration, certification status or other characteristic (including whether or not the materiel has been used previously) has been falsely represented by:

- misleading marking of the materiel, labelling or packaging;
- misleading documentation; or
- any other means, including failing to disclose information;

except where it has been demonstrated that the misrepresentation was not the result of dishonesty by a supplier or sub-supplier within the supply chain

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Note 1 to entry: Regardless of the governing law of the supply contract, the term “origin” is to be accorded the same meaning as the equivalent term found in European Directive 2008/95/EC (the “Trade Marks” Directive), and the terms “falsely represented”, “misleading”, “failing to disclose information” and “dishonesty” are to be accorded the same meaning as the equivalent terms found in the United Kingdom’s Fraud Act 2006.

Note 2 to entry: Within this document ‘materiel’ is referred to as ‘items’.

[SOURCE: DEF.STAN.05-135/1:2014 [16], modified – Note 2 has been added.]

A.6
bridge buy
lifetime buy for a given period, for example during the replacement item development

A.7
electronic components management plan
ECMP
plan owner's document that defines the processes and practices for applying components to an equipment or range of equipment and which generally addresses all relevant aspects of controlling components during system design, development, production, and post-production support

Note 1 to entry: This note applies to the French language only.

[SOURCE: IEC 62239-1:2018 [5], 3.1.18]

A.8
EOP forecast
predicted year the item production ceases to be made

Note 1 to entry: EOP forecast is also known as years to end of life (YTEOL).

A.9
franchised distributor
franchised agent
individual or corporate organization that is legally independent from the franchiser (in this case the electronic component manufacturer or OCM) and agrees under contract to distribute products using the franchiser’s name and sales network

Note 1 to entry: Distribution activities are carried out in accordance with standards set and controlled by the franchiser. Shipments against orders placed can be despatched either direct from the OCM or the franchised distributor or agent. In other words, the franchised distributor enters into contractual agreements with one or more electronic component manufacturers to distribute and sell the said components. Distribution agreements can be stipulated according to the following criteria: geographical area, type of clientele (avionics for example), maximum manufacturing lot size. Components sourced through this route are protected by the OCM’s warranty and supplied with full traceability.

[SOURCE: IEC TS 62668-1:2016 [11], 3.1.9]

A.10
obsolescence concern
future obsolescence issue that has been identified as a result of a proactive approach where a resolution needs to be developed and implemented to minimize the impact on future availability, and cost to, a project

Note 1 to entry: Within this document an "obsolescence concern" is referred to as an "obsolescence risk" (see 3.1.14).

[SOURCE: MoD Defence Logistics Framework [17]]

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A.11
original component manufacturer
OCM
manufacturer of a sub item, material or component that is intended for embodiment into an item, assembly or a product by an original equipment manufacturer (OEM)

Note 1 to entry: OCM is a common term used to identify an organization producing a sub item in the supply chain. This document now simply uses ‘manufacturer’ where ‘original component manufacturer’ refers to the sub item manufacturer identified in the original configuration of the item.

Note 2 to entry: This note applies to the French language only.

A.12
original equipment manufacturer
OEM
manufacturer of an item, assembly or a product

Note 1 to entry: OEM is a common term used to identify an organization producing an item in the supply chain. This document now simply uses ‘manufacturer’ where ‘original equipment manufacturer’ refers to the item manufacturer identified in the original configuration of the end user’s item.

Note 2 to entry: The assembly or product might be regarded as a component by a customer/end user.

Note 3 to entry: This note applies to the French language only.

A.13
recycled component
component removed from its original product or assembly and available for reuse

Note 1 to entry: The component has authentic logos, trademarks and markings. However, it typically has no output to measure the useful life remaining for its reuse. A recycled component can fail earlier than a new one when re-assembled into another product or assembly. A recycled component can also be physically or damaged through ESD during the removal process.

[SOURCE: IEC 62668-1:2019, 3.1.17 [11], modified — "electrical" has been deleted.]

A.14
risk mitigation buy
RMB
procurement of items sufficient to support the product throughout its life cycle, or until the next planned technology upgrade, to reduce an identified obsolescence risk to a project

Note 1 to entry: The risk mitigation buy is a proactive risk reduction measure, triggered by the user, where an unacceptable obsolescence risk to a project has been identified. Examples of RMB are: life time buy, life of type buy, bridge buy.

Note 2 to entry: Within this document a risk mitigation buy is a proactive approach of an LNB.

[SOURCE: MoD Defence Logistics Framework [17].]

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Annex B
(informative)

Obsolescence resolutions

B.1 Obsolescence management resolutions with EOP forecast

The scenarios for forecasting an EOP are illustrated in Figure B.1.

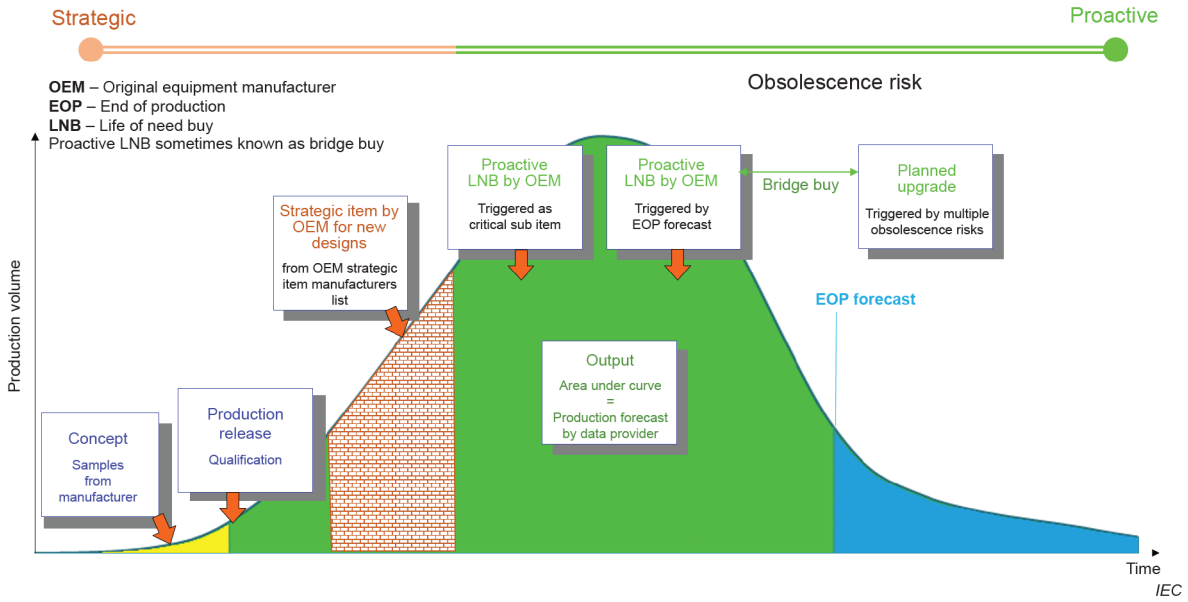


Figure B.1 – Item production output with EOP forecast and obsolescence scenarios

Following a configured list analysis, a sub item is identified as critical. This is as a result of the sub item being single sourced, being dependent on software and having an end-of-production forecast (EOP forecast) approaching. The result is a list of obsolescence risks that need resolution. These can be resolved either by stock management or by design strategies and resolutions.

NOTE The production output can represent the end user item, item (a line replacement unit, spares replacement unit or sub-assembly) or in this case a sub item.

Resolutions can take the form of a proactive LNB before the manufacturer has declared the sub item obsolete (also known as a risk mitigation buy).

Where an upgrade is planned it can be necessary to set-up a proactive LNB (also known as a bridge buy) until the design upgrade (also known as technology insertion) is assessed, qualified and the upgrade is implemented.

The implementation of the upgrade may take the form of:

- item emulation for a known EOP item, manufactured from an assessed aftermarket solution provider;
- design change or upgrade planned when there are EOP sub items forecast to be used in an item that has a long-term production life;
- design change for items planned when both critical and multiple EOP sub items are forecast and where there are software, firmware and interface dependency issues.

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B.2 Obsolescence management resolutions with EOP announcement

The obsolescence terms with an EOP announcement is as shown in Figure B.2.

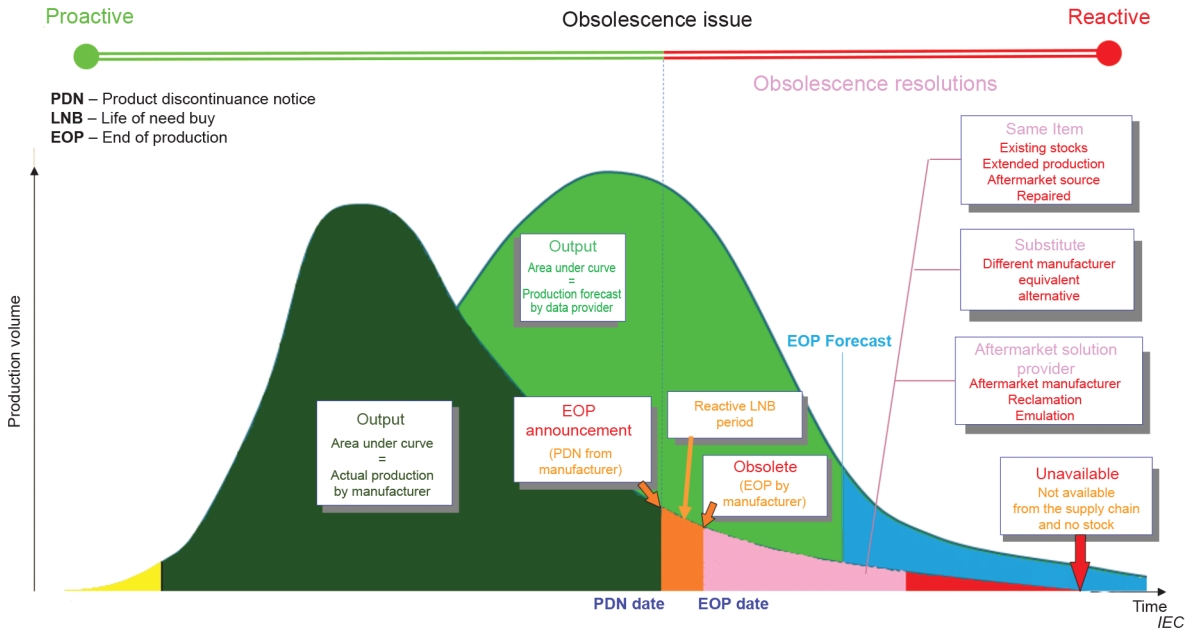


Figure B.2 – Item production output at EOP announcement and obsolescence scenarios

The sub item has been monitored by the item manufacturer for availability and eventually receives a PDN for the sub item. In the diagram above, the EOP forecast is later than the EOP date, overlaid on the previous 'production cycle' diagram.

Resolutions include:

- using existing stock, which was procured as a proactive LNB made for a critical sub item;
- processing a reactive LNB (can be known as a last time buy) from the sub item manufacturer;
- searching for an authorized aftermarket manufacturer in order to avoid counterfeit items;

  NOTE The date that is important to counterfeit avoidance is to record the 'last date of manufacture' when checking date codes on suspect counterfeit items.

- reclamation.

There might be design resolutions already in progress in the form of planned upgrades, or there can be substitutes:

- equivalent items;
- alternative items.

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B.3 Alternate manufacturers: example of a reactive approach in electronics

When attempts are made to procure an item, but the item has a PDN or EOP announcement or is already identified as obsolete by the item's manufacturer, an LNB is typically enacted by the organization, whereby a search is made for traceable stock. The dangers of this reactive approach are the risk that only untraceable stock can be found from brokers outside the franchised supply chain which might supply counterfeit or fraudulent recycled components. Further guidance on avoiding counterfeit and fraudulent components can be found in IEC 62668-1 [11] and IEC 62668-2 [14]. Another problem might be that insufficient stock is available for purchase for future spares and repairs. Some LNB stock also requires long term storage incurring additional storage costs (see IEC 62435-4 [13]).

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Annex C
(informative)

Guidance on the effects of obsolescence

Obsolescence can affect several types of items in various ways, for example:

- Electronic items: At electronic components level, obsolescence occurs regularly due to: increase in functional speed, reduction in features such as size and supply voltage, changes in materials (see bullet below for details) and in interconnection and packaging technologies. As a result, many of the electronic sub items (i.e. components) that compose a sub item have a life cycle that is significantly shorter than the life cycle of the item (e.g. a printed circuit assembly).
- Mechanical items: The item life of mechanical sub items can be much longer than electronic sub items, but obsolescence problems of mechanical sub items can escalate as the item they support starts to age. Failure of mechanical sub items can trigger obsolescence due to the potential unavailability of the item because of the unavailability of spares and materials.
- Materials can become obsolete due to the introduction of new environmental legislations and regulations such as the RoHS or REACH. Materials declared obsolete by such triggers can be incorporated into an item, possibly at a lower level in the supply chain. For example, a new type of plating chemistry is banned and will affect other things when changed. (See also BS EN 9278 [15].)
- Processes: Changes in environmental regulations are common drivers of obsolescence in manufacturing processes. In the light of this, a material obsolescence issue can make a manufacturing process obsolete and also the obsolescence of a manufacturing process can prevent the manufacture of an item or material, making it obsolete.
- Software obsolescence can occur when either:
  - hardware changes (whether due to obsolescence or upgrades) can make the software functionality obsolete;
  - the sales, support and/or maintenance for software stops as the manufacturer no longer sells the software and there is an inability to expand or renew licensing agreements;
  - external factors change, which then can request enhancements or operating environment change, security protection and additional software test, etc.
- Media: Digital media obsolescence: size, formatting types, or degradation limits or end of access to software or other technical information held.
- Manufacturing tooling: The manufacturing aids required to fabricate items are regarded as ‘tooling’ (for example: forging dies, holding fixtures, sheet metal patterns, casting moulds, jigs). Obsolete tooling might need to be refurbished or recreated. Otherwise, it can impact the ability to manufacture a correct item.
- Support and test equipment can become obsolete because it can be a commercial item or special to a type of equipment that is no longer available.

NOTE Management of test equipment is considered as part of the configured list and managed in the same way as any other sub item.

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Annex D
(informative)

Guidance on the OMP

Additional guidance to 7.2 on the contents of the OMP is as follows:

- authorities and resources
  - compatibility with the customer's/end user's current support arrangements, where applicable;
  - processes for communication within the organization, between the organization and throughout the supply chain;
- scope
  - current life cycle phase of the item, and all phases of the life cycle relevant to the organization, including the possibility of unplanned repeat orders;
- objectives
  - sustainment of the item at the end of the production phase and the role of obsolescence management;
- strategies to minimize obsolescence during design
  - collaboration of the whole supply chain with obsolescence management objectives negotiated with manufacturers and their franchised distributors (see A.9) and their authorized aftermarket manufacturers (see A.1));
- monitoring and schedule
  - the level and periodicity of the monitoring required for critical and single source items will be dictated by the level of risk;
  - identification of how critical software will affect the obsolescence of the item and whether software will be monitored;
  - determination of whether materials used in the manufacture of the item will be monitored;
- obsolescence management activities determination and implementation
  - likelihood of sub items obsolescence impacting the item;
  - likelihood of obsolescence of sub items and items used for production or provision of services;
  - possibility of item life extension and enhancement;
- resolution selection and implementation
  - consideration of the need for item qualification due to the implementation of a resolution;

    EXAMPLE 1 Using an electronic component management plan ECMP such as IEC 62239-1 [5] for electronic components.
- OMP performance
  - maintenance of the plan;
- tools used
  - configuration and change management.

    EXAMPLE 2 Item status data provider(s), databases such as an obsolescence data repository (ODR).

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Annex E
(informative)

Examples of an obsolescence risk assessment

E.1 General

There is uncertainty as to when obsolescence will have an impact, therefore the likelihood of an impact is characterized in terms of time, for example the probability of obsolescence having an impact within a specified time period or the point in time when obsolescence is predicted to have an impact. The aim of obsolescence risk assessment is to allow decision makers to know whether there is a low likelihood but very high impact or vice versa, and record the assumptions and data used to make the assessment. The time period for the likelihood assessment is defined in accordance with the period of interest for example between 0 and 5 years for a 5-year support contract or 0 and 25 years for a user managing a ship with a 25-year in-service life.

The two examples within Annex E describe this decision making process in three steps; however, this would not necessarily be required for simple systems.

E.2 Example 1

The likelihood assessment within Annex E determines the probability that an item will become obsolete within the period of interest, for example production or support period. The likelihood assessment shown in Table E.1 assesses the likelihood of an impact, therefore it takes account of the expected time to the item becoming obsolete and the predicted time when the stock will run out (current stock versus demand for the item). The time periods for the likelihood assessment align with the period of interest for example a support contract period of 7 years.

Table E.1 – Likelihood assessment

Years to item becoming obsolete	**< 3 years**	**Low**	**Medium**	**High**
	3 to 7 years	**Low**	**Medium**	**Medium**
	> 7 years	**Low**	**Low**	**Low**
		> 7 years	**3 to 7 years**	**< 3 years**
		Years to end of stock in hand		

Impact assessment as shown in Table E.2 takes account of the effect the unavailability of the sub item has on the item performance and/or procurement.

Table E.2 – Impact assessment

Impact on item performance	Impact on procurement	Criteria
Prevents item from being operated	Prevents critical milestone being met in procurement.	High
Item operational but degraded	Procurement shortfall requires resolution by the first maintenance period	Medium
Other impacts or non-critical items		Low

The obsolescence risk is assessed based on the combination of the likelihood and the impact as shown in Table E.3.

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Table E.3 – Combination of likelihood and impact assessment

Likelihood	Impact		
	Low	**Medium**	**High**
High	Medium	High	High
Medium	Low	Medium	High
Low	Low	Low	Medium

The outcome of the likelihood and impact assessment above determines whether a reactive approach is appropriate or if a proactive approach is required, as follows:

- low = reactive approach adequate (no further action required);
- medium/high = proactive approach required.

A proactive approach aims to reduce the likelihood of occurrence of obsolescence and/or decrease the impact when obsolescence occurs. Proactive activities should be proportionate to the predicted resolution cost and the time it will take to resolve. Selecting suitable activities requires the complexity of the potential obsolescence issue to be assessed, which in Table E.4 considers a combination of the level of item complexity, level of integration and qualification requirements.

Descriptions of the different levels of proactive approach are as follows:

- Low: undertake the groundwork to identify and prepare for obsolescence occurrence and monitor the item for possible obsolescence if resources permit.
- Medium: develop an action plan to ensure continued support, for example instigate LNBs.
- High: implement an aggressive approach by either designing out the problem or ensuring early implementation of resolutions that ensure continued support.

Table E.4 – Level of proactive approach assessment

		Integration/qualification requirements		
		Low	Medium	High
Item complexity	High - single source	Medium	High	High
	Medium - readily available, but will require some design changes	Low	Medium	High
	Low - multiple sources	Low	Low	Medium

NOTE 1 The level of integration describes the level with which an item is combined with another to form a whole (which could be the sub item or a higher level item). ‘Low’ would equate to a stand-alone item versus ‘high’ which would be an item that passes services (e.g. power) and data across the boundary with interfacing items.

NOTE 2 The level of qualification describes the complexity of the testing required to evaluate if a replacement sub item is acceptable. ‘Low’ would equate to a stand-alone test versus ‘high’ that would require the whole item to be tested.

E.3 Example 2

E.3.1 Risk assessment process

The process within this example fundamentally answers the following questions (see Figure E.1):

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- What is the likelihood of an item becoming obsolete?
- If it does become obsolete, when will this impact on the capability?



Figure E.1 – Sample risk assessment process

E.3.2 Likelihood assessment

The likelihood of an item becoming obsolete is the combination of the EOP forecast of the item, and the number of approved manufacturers that are currently available on the item's configuration (see Table E.5 and Table E.6).

Table E.5 – EOP forecast

EOP forecast	Assessment
Less than 2 years	High
Between 2 and 5 years	Medium
Greater than 5 years	Low

Table E.6 – Number of approved manufacturers

Number of approved manufacturers
None or one
Two
Three or more

The likelihood is assessed on the combination of the EOP forecast and number of approved manufacturers (see Table E.7).

Table E.7 – Likelihood

		Number of approved manufacturers		
		> 2	2	0 to 1
EOP forecast	< 2 years	Medium	High	High
	2 to 5 years	Low	Medium	High
	> 5 years	Low	Low	Medium

E.3.3 Impact date assessment

If an obsolescence issue arises, it is important to understand the date at which the obsolescence issue will have a direct detrimental effect on the operational capability of the equipment.

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This is done by comparing the current inventory level against the consumption rate of the item.

Impact months = (stock + stock on order) / consumption rate

Impact date = today's date + impact months

The impact date is classified as shown in Table E.8:

Table E.8 – Impact date

Impact date (from today)	Impact assessment
> 5 years	Low
2 to 5 years	Medium
Within 2 years	High

E.3.4 Obsolescence risk

Obsolescence risk is assessed based on a combination of the likelihood and the impact date (see Table E.9).

Table E.9 – Risk level

Probability	High	Medium	High	High
	Medium	Low	Medium	High
	Low	Low	Low	Medium
		Low	Medium	High
		Impact		

E.3.5 Review

The obsolescence risk assessment will be periodically reviewed and updated where necessary. Examples of periodicities are shown below (see Table E.10), these will depend on the support position at the time and the resources available.

All obsolescence risk review periods should be annotated in the OMP.

Table E.10 – Obsolescence risk review

Obsolescence risk	Period of review
High	Every six months
Medium	Annual
Low	Every two years

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Annex F
(informative)

Example of an obsolescence management decision process

The process described in Annex F starts with receiving information indicated in this case by active monitoring of the market, but it could equally be a PCN or any other form of market information. It is assumed obsolescence management experts will receive this information for action.

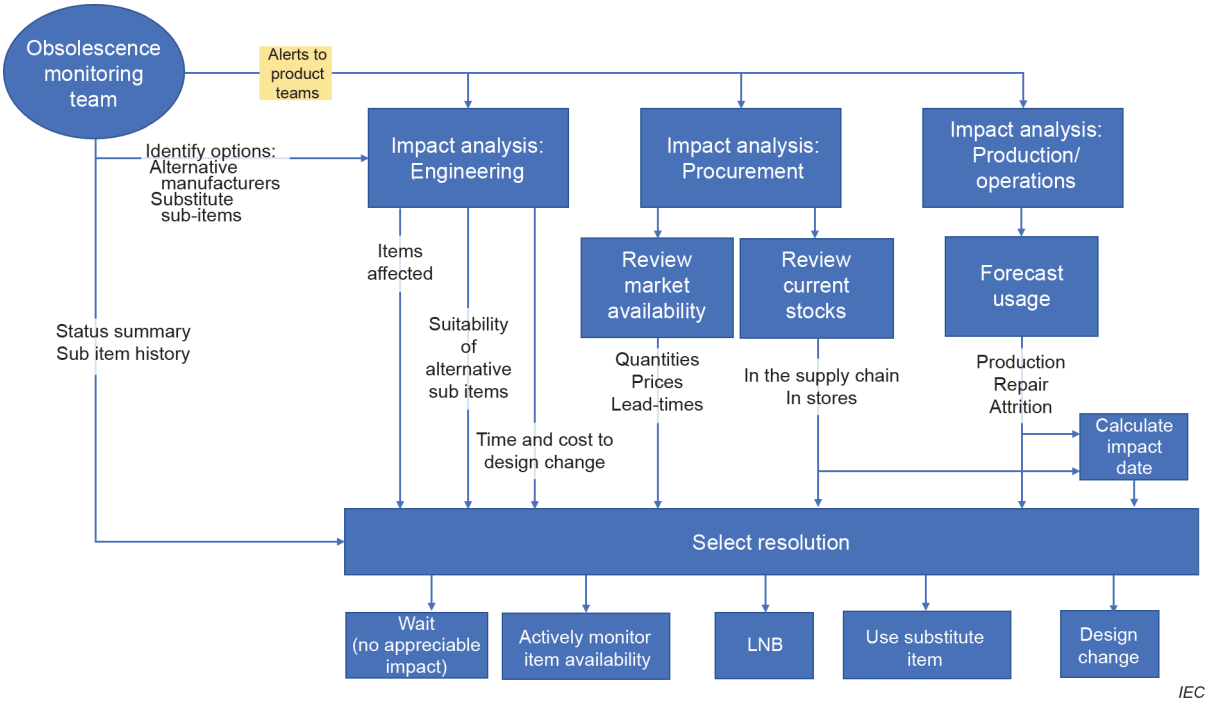


Figure F.1 – Initial decision process to recommend obsolescence resolutions

The process shown in Figure F.1 relies on decisions being made as part of a process to actively manage obsolescence in a structured way; just part of the process could be used or fewer actors could be involved.

Other resolutions than those indicated here can be used, these are the most common resolutions.

“Impact” dates are when there will be a definite impact on the item, not a predicted impact.

Although these flow charts indicate single resolutions, multiple resolutions may be needed, depending on time and availability, for instance, a short term LNB may be needed to reach a long term design change point.

This process may need constant revisiting until the issues identified are resolved; good record keeping (case notes) is essential in achieving this aim.

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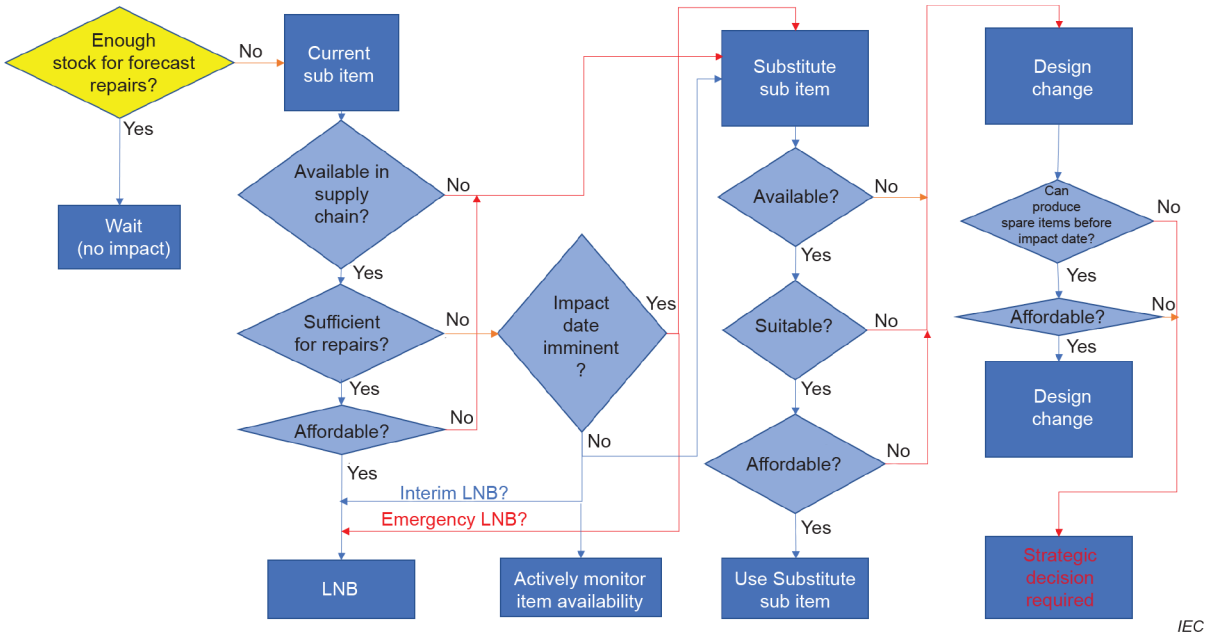


Figure F.2 – Decision process to recommend obsolescence resolutions (long term repairs strategy)

Figure F.2 and Figure F.3 show further analysis of the situation will be required after the initial decision to proceed has been taken.

NOTE In Figure F.2 and Figure F.3 lines have varied colours and weighting to clarify the paths available.

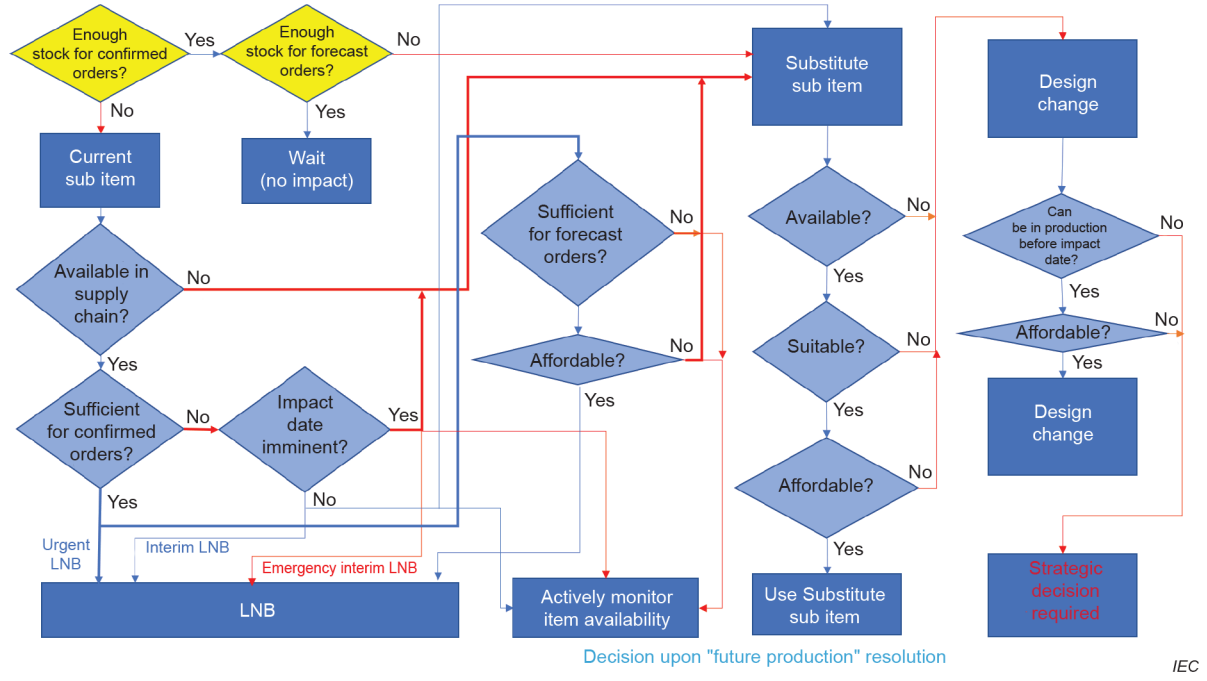


Figure F.3 – Decision process to recommend obsolescence resolutions (LNB)

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